Phase 2 · Chunk 1 Status Report

Agent harness evaluation on real SCG tickets · prepared for Rohit · 2026-11-05

STATUS: PAUSED · waiting on the Anthropic API spend limit

50%	9	$12.83	3
complete (9 of 18 jobs)	jobs validated and saved	spent of the $50 budget	harnesses compared

Evaluation progress — **9 / 18 jobs**

Budget used — **$12.83 / $50.00 (26%)**

Job tracker

Ticket	Type	pi	opencode	mini-swe-agent
SCG-140	bug	**PASS**	**PASS**	**PASS**
SCG-141	feature	**PASS**	**PASS**	**PASS**
SCG-137	feature (v1 text)	**FAIL**	**FAIL**	**FAIL**
SCG-139	bug	**INVALIDATED · rerun**	**INVALIDATED · rerun**	**INVALIDATED · rerun**
SCG-133	bug	**PENDING**	**PENDING**	**PENDING**
SCG-137	feature (v2 text)	**PENDING**	**PENDING**	**PENDING**

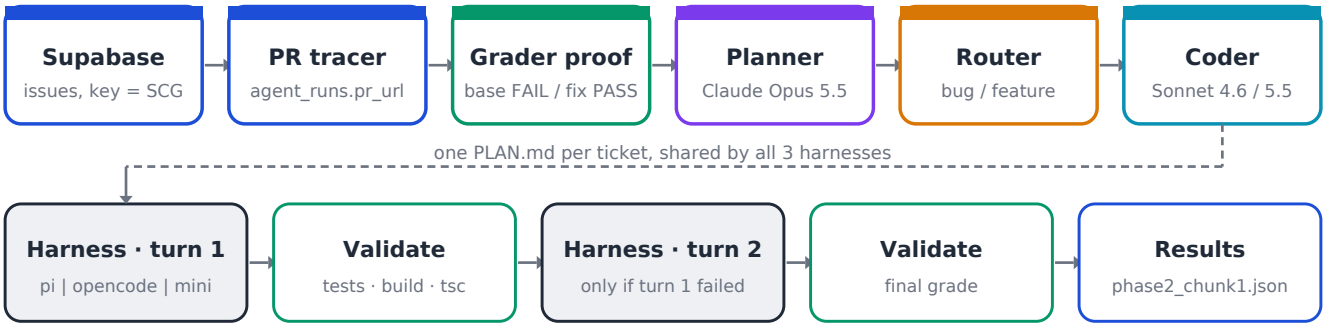
PASS/FAIL = final result after at most 2 turns · INVALIDATED = run without isolation, its result was thrown out · PENDING = waiting for API access

Key takeaways

- **All 9 finished jobs are validated.** Each was graded on the real PR's hidden tests, 1,300–1,800 regression tests, the web build and a baseline-relative typecheck.
- **Accuracy is tied** (2/3 each). **pi is the cheapest and fastest**: $0.40 in total, compared with $2.07 for OpenCode.
- **An isolation gap was found and closed.** Agents could read sibling worktrees. A 4-layer rootless sandbox now blocks this, verified in a free test.
- **Blocker:** the Anthropic key reached its custom spend limit. Raise it in the Anthropic Console (Settings → Limits) to finish the remaining 9 jobs (about 2–2.5 h, about $8–12).

Architecture & Configuration

Pipeline



Two-step model router

Step	Model	Price (per 1M tokens, in / out)	Role
1 · Planner	**claude-opus-5-5**	$4 / $20	Reads the repo and writes a strict PLAN.md (5 plans, $1.43 total)
2 · Coder (bug, small_task)	**claude-sonnet-4-6**	$3 / $15	The harness implements the plan in its own workspace
2 · Coder (feature, big_task)	**claude-sonnet-5-5**	$2 / $10	Same, for multi-file feature work

Agent harnesses

Harness	Version	How it runs	Turn 2
pi	0.84.3	pi -p --mode json (built-in Anthropic, custom 5.5 models)	resumes the same session
opencode	1.19.5	opencode run --format json --auto	--continue the same session
mini-swe-agent	2.4.6 (litellm 1.108)	mini -y --exit-immediately, its own $ limit	new run + failure log + plan

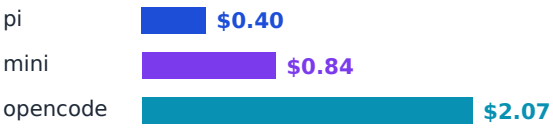
Data pipeline & grading

#	Stage	Detail	Result
1	Fetch tickets	Supabase REST (secret key, GET only): `projects.key=eq.SCG`, state DONE, repo scg-cash-management	91 issues
2	Trace the fix	`agent_runs.pr_url` → squash commit `(#N)` on main → base = its parent	38 mapped
3	Hidden tests	The PR's test files. Fail-to-pass and pass-to-pass sets proven on base vs. the real fix	5 graders proven
4	Route	Keyword scores + the `complexity` field + fix size → bug / feature	3 bugs · 2 features
5	Baseline	Full web + app suites, web build and tsc on clean code, once per ticket	0 failing tests
6	Grade a turn	Hidden tests pass, nothing that passed breaks, build OK, no new tsc errors, ≥ 1 file changed	PASS / FAIL

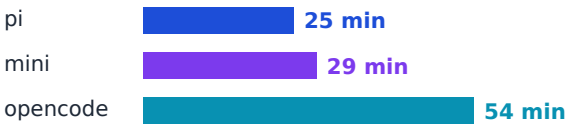
Performance Metrics · 9 validated jobs

Harness	Final pass	Turn-1 pass	Total cost	Cost / job	Wall time	Cache tokens read	Verdict
pi	2 / 3	2 / 3	$0.40	$0.13	25 min	0.37 M	**cheapest · fastest**
mini	2 / 3	2 / 3	$0.84	$0.28	29 min	1.11 M	middle
opencode	2 / 3	2 / 3	$2.07	$0.69	54 min	3.86 M	**most expensive**

Total cost (USD)



Total wall time (minutes)



Results matrix (cost per job)

Ticket	Type	Coder model	pi	opencode	mini-swe-agent
SCG-140	bug	sonnet-4-6	**PASS · $0.15 · 1 turn**	**PASS · $0.68 · 1 turn**	**PASS · $0.27 · 1 turn**
SCG-141	feature	sonnet-5-5	**PASS · $0.09 · 1 turn**	**PASS · $0.14 · 1 turn**	**PASS · $0.14 · 1 turn**
SCG-137 (v1)	feature	sonnet-5-5	**FAIL · $0.17 · 2 turns**	**FAIL · $1.26 · 2 turns**	**FAIL · $0.43 · 2 turns**

What the numbers say

- **Same model, different harness, about 5× cost gap.** OpenCode read 3.86 M cached tokens against pi's 0.37 M for the same 3 tickets.
- **Every validated job kept quality:** 0 broken regression tests, the build was never broken, and no new typecheck errors.
- **SCG-137 v1 failed for all 3 harnesses, for one reason.** The hidden test looks for exact labels (`Remove <id> from per diem`) that the ticket never names. A v2 rerun with an interface note is queued.
- **The planner is a single point of failure.** On SCG-139, the Opus plan pointed at the web form, but the real fix is in the printed form. Every harness followed it.

Incident & Fix · the Rootless Sandbox

What happened (SCG-139, 22:14–22:50)

Harness	Turn	Behaviour seen in the logs	Impact
pi	2	After "tests still fail": `ls eval-worktrees/phase2/`, read OpenCode's test files and copied them to /tmp	result invalid
opencode	1–2	`find` across the whole research folder, listed the SCG-139-base worktree	result invalid
mini	1	Worked in a Phase 1 worktree (`eval-worktrees/ui-b1716e69`), not its own	result invalid
all	—	No access to .env, tokens, keys or ~/.ssh (every log scanned). The main repo was untouched. The Phase 1 worktree is restored.	no secret exposure

Root cause: every job had its own git worktree, but harnesses run as the same Linux user. Sibling worktrees, results and history were all readable. The 9 earlier jobs were checked and showed **no access outside their own worktree**.

The fix: 4-layer rootless sandbox (no sudo, no user namespaces on this VM)

1 — Isolated /tmp workspace
git archive of the base commit + hard-linked node_modules + a fresh 1-commit git repo. '..' only reaches /tmp, and git log shows no future commits or PRs.

2 — Command guards (PATH)
chmod blocked. grep, rg and find refuse '..' and /home/mukesh paths. ls, cat, head and tail refuse /home/mukesh paths.

3 — Directory lock
Research folder chmod 0000 while a harness runs, then restored to 775 (also on crash or exit). Private HOME and XDG dirs.

4 — Prompt sanitization
Turn-2 feedback lists failing checks as plain assertions. No mention of "hidden" or "acceptance" tests.

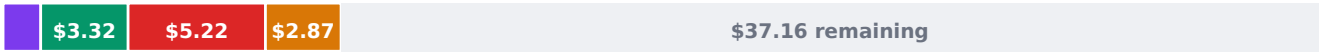
Verification (free test, research folder locked, no model calls)

Escape attempt	Result	Normal work inside the sandbox	Result
cat .env (guard, then /usr/bin/cat)	blocked · denied	vitest suite file	29 / 29 pass
ls the research folder / a Phase 1 worktree	Permission denied	web build (vite)	built 10.7 s
find .. / chmod 755	blocked	backend test (tsx --test)	3 / 3 pass
git log / branches / remotes	1 commit · main only	pi · opencode · mini startup	reach Anthropic

Limit: same-uid hardening stops agents from wandering, not a determined attacker. A container (Docker or bubblewrap, which need sudo) remains the long-term fix. With the global lock, jobs run one at a time.

Budget & Engineering Log

Where the $12.83 went



Planner (5 plans) · 9 validated jobs · SCG-139 runs (invalidated) · Reruns & interrupted turns

Bar scale = the full $50 budget

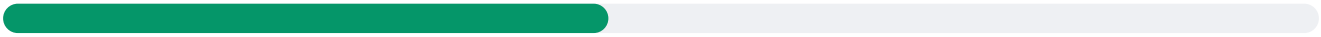
Issues found and fixed during the run

#	Issue	Effect if left	Fix	Status
1	JUnit parser: an unescaped > in a test name	Lost ~65 tests per baseline; false regressions	Quote-aware parser + re-parse of saved reports	fixed
2	tsc errors contain each worktree's absolute path	Phantom "+1 new error" failed every turn 1	Normalize worktree paths before comparing	fixed
3	mini first-run setup wizard + missing mcp package	mini never reached the model ($0 runs)	MSWEA_CONFIGURED + pip install mcp	fixed
4	Stopping left validation scopes running	Orphan test runs used CPU	Track and kill every scope, also on crash	fixed
5	Agents could read sibling worktrees	Invalid results (SCG-139)	4-layer rootless sandbox	fixed · tested
6	Hidden tests check strings the ticket never names	Unfair fails (SCG-137 v1)	v2 rerun with an interface note	queued

Next steps

	Step	Owner	Time	Cost
☐	**Raise the Anthropic API spend limit** in the Anthropic Console (Settings → Limits). It otherwise resets on 2026-12-01	Rohit / account admin	minutes	—
☐	Resume the sandboxed run: SCG-139 × 3 and SCG-133 × 3 (one command, detached)	Engineering	~1.3 h	~$5–7
☐	SCG-137 v2 rerun × 3 with the interface note (starts automatically after the step above)	Engineering	~0.7 h	~$2–4
☐	Final report: 18 jobs, harness ranking and a recommendation for the agent-core Worker	Engineering	~0.5 h	$0
☐	Make the GitHub research repo private, then commit Phase 2 code and results	Repo owner	minutes	—

Projected budget at completion **about $21–25 of $50**



Artifacts

Path	Contents
`eval-results/phase2_chunk1.json`	9 validated jobs, 3 invalidated (with reasons), 9 not run, budget, plans
`phase2/`	Orchestrator, router, validator, harness adapters, sandbox (`src/sandbox.mjs`)
`~/.local/share/p2-sbx/run-remaining.sh`	One-command resume for the remaining 9 jobs
`eval-results/phase2/chunk1.log`	Full run log, including every turn result and every incident